

PART B – Sourcing Strategy + Scale-Up Proposal

Question 1: Sourcing Methods

S. no.	Method	Why It Works for Federer ICP	Limitations
1	DSIR Recognised R&D Units List (govt)	Direct evidence of C3 (R&D investment) and C4 (technical leadership). DSIR certification means the company has in-house R&D, a strong differentiator.	Skews to older, established companies; misses new startups.
2	USFDA / EU-GMP / WHO-GMP Approved Facility Lists	Regulatory approval = strong C3 (differentiation) + C5 (export tailwind). Only serious manufacturers pursue these.	Heavily pharma/chem biased; doesn't cover defence, agri, medtech.
3	State Industrial Development Corporation Databases (TSIIC, MIDC, GIDC, KSIDC, etc.)	These list companies that have been allotted land, power connections, or industrial plots – all real manufacturers, not traders.	Data is often messy, incomplete, or offline. Requires manual cleaning.
4	Trade Expo Exhibitor Directories (CPHI, BioAsia, Aero India, IMTEX, ELECRAMA, PackEx, Agri Intex)	Exhibitors pay ₹2-10 lakh for a booth – a growth signal (C6). Segments map directly to our industry baskets.	Biased toward companies that can afford booths; misses bootstrapped MSMEs.
5	Industry Association Member Directories (IDMA, BDMA, ELCINA, ACMA, Seed Association of India, TMMA, etc.)	Active membership signals an operating company. Many directories include size, products, and sometimes revenue.	Some are behind paywalls; data quality varies.
6	BSE SME / NSE Emerge Listed Companies	Public financials (revenue band, leadership names, annual	Only captures listed companies; misses

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		reports) make scoring fast and reliable.	unlisted private MSMEs (which are the majority).
7	MCA (Ministry of Corporate Affairs) Data via Antigravity/Tofler	NIC codes let you filter by manufacturing sub-segments. Director details give decision-maker names.	Very noisy – NIC codes are self-reported and often wrong. No website data.
8	LinkedIn Sales Navigator (Inverted Search)	Search for founders/MDs with technical degrees (PhD, IIT, IISc) in target cities, then map to their companies. Excellent for C4.	LinkedIn data self-reported; many MSME leaders don't have updated profiles.
9	Customs Import Data (e.g., Zaubas, ImportGenius)	Identify companies importing specialised machinery (German/Japanese CNC, bioreactors, Swiss textile machines). High-value capital goods imports = capability-based differentiation (C3).	Data access costs money; requires interpretation (not all importers are manufacturers).
10	Patent Databases (Indian Patent Office, WIPO)	Search for patent assignees located in target cities. Patents = strong C3 (IP differentiation).	Many patents are from large R&D centres, not MSMEs. Still a useful filter.

Question 2: 1000-Company in One Month

Executive Summary

I will build a verified list of **1,000 companies** matching DeepThought's Federer ICP (₹50-500Cr, promoter-driven, differentiated product, technical DM, active growth signals) within **30 calendar days** using a combination of public data sources, AI-assisted scoring, and targeted human quality assurance. The expected yield is ~30%, so I will source an initial universe of **3,500-4,000 companies**.

Tools & Licences:

- **Claude API** (Haiku for first-pass, Sonnet for borderline re-scoring)
- **Gemini API** (alternative for cross-checking)
- **GitHub Copilot** (code assistance)
- **Antigravity** (MCA data extraction)
- **Playwright + Python** (web scraping)
- **LinkedIn Sales Navigator** (DM discovery)
- **Google Sheets / PostgreSQL** (data storage)
- **Residential proxies** (if scraping gets blocked)

Week-by-Week Plan

Week 1 – Build the Universe (3,500-4,000 companies)

Day	Task	Output
1	Download & parse DSIR list, BSE SME / NSE Emerge lists, USFDA/EU-GMP facility lists.	Structured CSVs.
2	Scrape expo exhibitor directories (CPHI, BioAsia, Aero India, IMTEX, ELECRAMA, PackEx, Agri Intex). Use Python + Playwright with 10s timeout.	600-800 names.
3	Extract industry association directories (IDMA, ELCINA, ACMA, Seed Association).	500-600 names.
4	Query MCA data via Antigravity – NIC codes 20-29 (manufacturing), paid-up capital > ₹1Cr, cities target list.	1,000-1,500 raw entries.
5	LinkedIn Sales Navigator extraction – search for founders/MDs in target cities + technical keywords (IIT, PhD, ex-ISRO, etc.).	400-600 profiles → ~300 companies.
6	Deduplication – fuzzy matching on company name + city. Use thefuzz library.	3,500-4,000 unique companies.

Day	Task	Output
7	Website discovery – for entries without URL, automate Google search: "{company name} {city} site:co.in OR site:com".	≥90% have a website.

Week 1 output: Master list with columns: Company Name, City, Segment (inferred), Website (if found).

Week 2 – Automated Pre-filter & First-Pass AI Scoring

Day	Task	Details
8	Hard pre-filter (no AI) – reject if: no website, parked page, keywords “trading/distributor/CRO/testing lab”, known revenue >₹500Cr (from BSE).	Reduces 3,800 → ~2,500.
9	Build scraper – for each remaining company, scrape: /, /about, /leadership, /products, /news, /careers. Concatenate to ~8K tokens.	Scraper tested on 50 companies.
10	Write scoring prompt for Claude Haiku – output JSON with C3-C8 scores (0, half, full) + evidence + confidence flags.	Prompt calibrated on 15 test companies.
11-13	Run scraping (parallel, 4 workers, 30 sec/company).	All 2,500 companies scraped.
14	Run Haiku scoring (3 sec/company). Cost: ~₹0.40 × 2,500 = ₹1,000.	First-pass results: ~1,200 passes (score >60).

First-pass yield: ~1,200 companies with score >60.

Week 3 – Borderline Re-scoring & Quality Control

Day	Task	Details
15	Identify borderline & low-confidence – all companies with total score 40-60, or any confidence flag = “low”.	Expected ~700 companies.
16-17	Re-score with Claude Sonnet (more accurate, 5 sec/company). Cost: ₹3.5 × 700 = ₹2,450.	Improved scores.
18	Programmatic QA – flag false patterns: C3 evidence only “ISO 9001”, C6 evidence older than 2023, C4 evidence generic “ B.Tech ” without institution.	Flagged list ~300-400.
19-20	Human QA – manual review of flagged companies (3-4 min each = ~20 hours). Verify 2-3 evidence points per company. Adjust scores, reject false positives.	Final passes after QA: ~1,000.

Quality control rules enforced:

- **C3:** “Strong” requires patents, DSIR, USFDA/EU-GMP, OR specialised equipment + custom capability. ISO 9001 alone = Moderate at best.
- **C4:** “Strong” requires PhD/IIT/IISc/ex-ISRO/DRDO, OR documented systems thinking (ERP, costing models). [B.Tech](#) from unknown college = Moderate.
- **C6:** Must have ≥2 signals from hiring, facility, certification, website currency, financial growth. News older than 2023 = 0.

Week 4 – Final Assembly & Enrichment

Day	Task	Details
21-22	Merge all passes from clean strong_pass + QA-verified pass. Deduplicate again.	Target 1,000 companies.
23-24	Add personalisation hooks for top 200 priority companies – one specific, recent, true detail per company (e.g., “new facility inaugurated Oct 2025”, “first	Ready for outreach.

	Indian company to get X approval”).	
25-26	Format final CSV – columns as per assignment: Company Name, Website, City, Segment, What They Make, Revenue Band, Decision Maker, DM Title, DM Background, C1-C6 scores + evidence, Federer Score, Score Band, Verdict, Verdict Reasoning, Personalisation Hook, Evidence Sources.	Final deliverable.
27-28	Write methodology document – describe sources, filters, yield rates, QA process, AI prompts used, limitations.	Transparency for client.
29	Package code – all scraping scripts, scoring notebooks, prompts, data processing steps.	Reproducibility.
30	Buffer day – handle overruns, re-run any failed scrapes.	Contingency.

Realistic Yield Table

Stage	Input	Output	Cumulative Yield
Raw universe (from 7+ sources)	3,800	–	–
After hard pre-filter (no AI)	3,800	2,500	66%
After Haiku scoring (first-pass)	2,500	1,200	48%
After Sonnet re-score + Human QA	1,200	1,000	83% (of first-pass)

Final yield from raw universe: $1,000 / 3,800 = 26\%$ (within expected 25-30%).

Cost Estimate (One-Time)

Item	Cost (₹)
Claude Haiku API (2,500 companies)	₹1,000
Claude Sonnet API (700 borderline)	₹2,450
Residential proxies (if needed)	₹1,000
Total	~₹4,500

All other tools (Playwright, Python, GitHub Copilot, Antigravity, LinkedIn Sales Navigator) are assumed provided.

Risk Mitigation

Risk	Mitigation
Lower yield than 26%	Expand universe: add state industrial directories, Google Maps scraping of industrial areas, more expos.
Scraper blocked	Use rotating user agents, delays (1-2 sec), residential proxies for critical sites. Fallback to manual for top 500.
AI hallucination	Two-layer verification: programmatic flagging + human QA on 30-40% of passes. Evidence must be specific and verifiable.
Too many borderline cases	Tighten pre-filter rules; increase weight of C6 (growth signals) to favour recent activity.
Duplicates	Fuzzy matching on company name + city; manual review of top 100 suspect pairs.

Final Deliverables (End of Month)

1. Master CSV – 1,000 companies with all required columns.
2. Priority-200 CSV – top 200 with personalisation hooks.
3. Methodology document – detailed explanation of sources, filters, AI prompts, QA process, yield.
4. Code repository – all scripts, notebooks, and prompts (public or private GitHub).
5. Dashboard – simple visualisation of passes by city, segment, and score band (optional but impressive).

Personal Touch & Improvements

1. Unconventional Source Specific to Hyderabad (My Chosen City)

In addition to the seven core sources listed in the main proposal, I would add one Hyderabad-specific unconventional source:

T-Hub Incubator & Accelerator Alumni List

- T-Hub is India's largest startup incubator, based in Hyderabad. It has nurtured hundreds of deep-tech hardware, medtech, and advanced manufacturing startups over the last 5-7 years.
- Why it fits the Federer ICP: Many of these alumni have now graduated from the prototype stage to commercial manufacturing, with revenues typically in the ₹10-50Cr range. They are technically driven (founders often from IIT/IIIT Hyderabad), have differentiated products, and are actively growing. They are ideal "emerging Federer" companies that could scale into our ₹50-500Cr band within 1-2 years.
- How to extract: T-Hub publishes annual reports and alumni directories. I would scrape these, filter for "manufacturing", "hardware", "biotech", "defence tech" categories, and then score them normally.
- Limitation: Many are still below ₹50Cr – but they are high-potential future targets and can be nurtured.

2. Lessons Learned from Part A (Hyderabad Pharma Companies)

From manually scoring the 14 Hyderabad-based pharma companies in Part A, I identified key insights that will directly improve the Part B process:

Lesson	Implication for Part B (1000-company plan)
Even companies with PhD founders can be disqualified – e.g., Divi's, Laurus, MSN all have brilliant scientist-founders, but their revenue (>₹500Cr) puts them outside our ICP.	Pre-filter strictly by revenue first using BSE SME / NSE Emerge data before any AI scoring. This avoids wasting API calls and time on large companies.
Founder's technical degree alone is not enough for C4 – Vivimed Labs' founder has a PhD, but the company is in financial distress (C6 weak).	Require both technical background AND at least one growth signal (C6) for a "Strong" C4. No growth = no pass, regardless of degree.
Regulatory approvals (USFDA, EU-GMP) are strong C3 evidence but don't guarantee size – Vasudha Pharma has multiple global approvals and is within revenue band.	Use regulatory approval lists (USFDA/EU) as a primary sourcing source – they already filter for quality manufacturers.
Service companies hide in plain sight – None of the 14 were service companies, but the sample CSV had Vimta Labs (CRO).	Add a hard keyword block for "analytical services", "testing lab", "CRO", "clinical research" before any scraping.

Operational change in the proposal:

In Week 2 (hard pre-filter), I will now add a revenue pre-check using BSE SME / MCA paid-up capital as a proxy, rejecting any company with estimated revenue >₹500Cr before scraping. This will save ~15-20% of API costs.

3. Simple Scoring Rubric for "Confidence" (to Reduce Hallucination)

To ensure that every score in the final list is trustworthy, I will add a confidence flag for each of the six criteria (C3-C8). The flag will be determined automatically based on evidence sources:

Confidence Level	Definition	Criteria
High	Evidence comes from two or more independent, verifiable sources (e.g., website + LinkedIn + news article).	C3: Patent number + DSIR listing. C4: LinkedIn profile + company leadership page.
Medium	Evidence comes from one reliable source (e.g., company website only) but is specific and current.	C6: “New facility announced on company news page (dated 2025)”.
Low	Evidence is vague (“seems to be growing”), from an unreliable source, or only inferred.	Any criterion where the LLM says “likely” or “appears”.

How this is usefull:

- Any company with two or more “Low” confidence flags will be automatically sent to human QA (even if the total score is high).
- The final CSV will include a column Overall Confidence (High/Medium/Low) based on the lowest confidence among C3-C8.
- Only companies with “Medium” or “High” overall confidence will be included in the final 1,000 list.

4. Handling Edge Cases – Companies Without a Website or With Missing Data

Not all qualifying companies have a complete web presence. I will handle edge cases with a tiered approach:

Edge Case	Action
No website found (after automated Google search)	If the company appears in DSIR, MCA, or USFDA lists (i.e., has a regulatory footprint), I will attempt a manual phone call to the registered number (from MCA) to verify manufacturing and request a website or basic details. Maximum 50 such calls (focused on high-potential segments).

Edge Case	Action
Website is a single parked page (e.g., “Under Construction”)	Same as above – fallback to MCA + phone call. If no response, the company is rejected.
No leadership page / decision-maker information	Use LinkedIn Sales Navigator to search for the company name and identify the founder/MD. If still not found, mark C4 as “Unknown” and auto-flag for human QA.
Revenue band unclear (neither BSE nor MCA data)	Use conservative estimate based on employee count (from LinkedIn or website). Rough proxy: 50-200 employees → ₹30-100Cr; 200-500 employees → ₹100-300Cr; >500 employees → >₹500Cr. If still unknown, mark as “<₹50Cr” and include only if other scores are very strong ($C3+C4 > 30$).

This matters because:

Some excellent Federer companies (especially older MSMEs) have minimal websites but strong fundamentals. This edge-case protocol ensures we don’t miss them, while still maintaining quality control.

5. Final Note – How These Improvements Connect to DeepThought’s Values

DeepThought looks for “fire in the belly” and curiosity, not just mechanical execution. By adding:

- A city-specific unconventional source (T-Hub),
- Lessons learned from real data (Part A),
- A structured confidence rubric (to fight AI hallucination), and
- Edge-case protocols (to avoid missing hidden gems),

I am demonstrating that I don’t just follow a template – I adapt, learn, and improve the process. This is exactly the mindset DeepThought wants: systematic but curious, data-driven but not blindly trusting automation.